

Module 4 Assignment: Open-Loop TEC Calibration And Software Safety

Purpose

Turn the manually controlled TEC from Module 3 into a measured open-loop process. You will measure how its steady-state temperature responds to PWM, analyze why heating and cooling differ, and add a software temperature limit while retaining the independent hardware thermal switch.

Learning Objectives

By the end of this module, you should be able to:

- operate the TEC safely using independent hardware and software protection;
- measure the steady-state temperature response to signed PWM and determine the heating and cooling susceptibilities;
- explain unequal heating and cooling slopes using Peltier transport, Joule heating, and passive conduction; and
- extract relevant manufacturer specifications from a data sheet and evaluate whether experimental measurements are consistent with them.

Before Class

1. Review your Module 3 Arduino sketch and Python GUI.
2. Confirm that you can set PWM and heat/cool direction manually.
3. Review how your Python program records or displays temperature versus time.
4. Read the [hardware page section on the thermal safety switch](#).
5. Read the [hardware page section on the TEC](#).

Outside-Class Workload Budget For S8

Work	Planned time
Read this assignment, review the safety boundary, and plan the data table	30 minutes
Make the calibration graph and extract its two slopes	15 minutes
Complete the guided A2 energy-balance analysis	30 minutes
Check and submit A2	15 minutes
Total outside class associated with S8	1 hour 30 minutes

All physical runs and safety tests occur in class. If analysis reveals that a measurement must be repeated, identify it for the next supervised opportunity rather than exceeding the four-hour outside-class limit.

Pre-Class Questions

1. What does it mean for the TEC/block temperature to reach steady state?
2. Why should you wait before recording a steady-state temperature?
3. Why might heating and cooling have different slopes in a plot of temperature versus PWM?
4. Why is a software temperature limit useful even when a hardware thermal switch is present?

Part 1: Prepare The Instrument And Add The Safety Interlock

Start from your working Module 3 setup.

Safety And Startup Checklist

Start with the actuator power supply turned off and disconnected. This current path was built in Module 3, but it must be inspected before calibration.

1. Confirm that 18 AWG stranded copper wire is used for every high-current connection:
2. power supply to H-bridge **B+** and **B-**,
3. H-bridge **M+** and **M-** to the TEC circuit, and
4. both wires connected to the thermal switch.

5. Confirm that both Module 3 female spade crimps remain secure and that a multimeter shows continuity through the closed thermal switch.
6. Confirm that the thermal switch remains in series with the TEC current path so opening the switch interrupts TEC current independently of the Arduino software.
7. Confirm that the H-bridge outputs were checked with TEC power off in Module 3.
8. Upload the Arduino sketch that receives PWM and heat/cool commands from Python, start the Python GUI, and confirm that PWM begins at 0.
9. Confirm that the displayed temperature is plausible.
10. Draw the complete high-current path in your notebook and ask the instructor to inspect the wire gauge, polarity, spade connections, thermal-switch placement, and power-supply current limit.

Do not enable actuator power until the instructor approves the completed wiring.

During this module, keep the measured temperature between **10 °C and 45 °C**. Stop the run if the temperature moves unexpectedly, the display freezes, the power-supply current rises unexpectedly, or the TEC or H-bridge becomes hot to the touch.

Add And Verify The Software Temperature Limit

Continue using the measurement sequence from Module 2: average 1000 raw thermistor-voltage measurements before calculating each temperature. Your data should update about once a second. If it is slower, something is wrong. This applies to the displayed temperature, recorded data, and software safety check.

Before collecting calibration data, modify the Arduino sketch so that it disables TEC PWM if the measured temperature exceeds **60 °C**. The hardware thermal switch opens near 70 °C and remains the independent final protection.

Your code should:

- define a named constant for the software temperature limit,
- check the averaged temperature every loop,
- set both H-bridge PWM outputs to zero when the limit is exceeded,

- continue printing serial data so the Python GUI shows what happened, and
- report clearly in the serial output when the safety shutdown is active.

Do not intentionally heat the apparatus to 60 °C. Set the software limit to about 30C and verify that the shutdown activates when you warm it up above that temperature and both PWM outputs are set to zero. Then restore the limit to 60 °C and show the result to the instructor.

Start The TEC

After the wiring and software interlock are approved:

1. Confirm again that PWM begins at and the temperature is plausible.
2. Enable the power supply.
3. At low PWM, test both heat and cool and confirm that the temperature responds plausibly. In the strip chart, the PWM trace should be red during heating and blue during cooling.

Record the Arduino sketch filename, Python filename, serial port, power-supply voltage, and power-supply current limit in your module notes. The latter values are written on the side of the power supply.

Part 2: Choose Direction And PWM Values

The Arduino treats **PWM as an 8-bit nonnegative magnitude** and uses a separate **1-bit heat/cool value** to select direction. Record both quantities for every measurement. The Python strip chart communicates the heat/cool bit visually by drawing the PWM trace red for heat and blue for cool.

Find one heating PWM value that produces **45 °C ± 2 °C at steady state**. Separately, find one cooling PWM value that produces **10 °C ± 1 °C at steady state**. Those two endpoint values define the maximum useful PWM magnitude for heating and cooling, respectively. The two maxima will generally differ.

For each direction, use five selected PWM magnitudes: 0%, approximately 25%, 50%, 75%, and 100% of **that direction's own** maximum useful PWM. Thus the heating set should extend from 0 to the PWM that gives 45 °C ± 2 °C at steady state, while the cooling set

should extend from 0 to the PWM that gives $10\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ at steady state. Record the exact integer values that you actually use.

Part 3: Measure Steady-State Temperature

Measure steady state only at the five selected heating values and the five selected cooling values from Part 2. Do not make a continuous 0–255 PWM sweep. In each direction, work through only the five chosen values, beginning at low PWM. After a sudden PWM change—for example, from 25% to 50%—the temperature approaches its new value approximately exponentially. It never becomes mathematically stationary: after one time constant it has completed about 63% of the change, after two it has completed about 86% (14% remains), and after three it has completed about 95% (5% remains).

Use the following practical definition of steady state for this lab. Estimate the time constant from the response to a PWM step, wait about three time constants, and then watch the trace for one additional minute. Because the temperature trace has noise, do not expect a perfectly horizontal line. Call the temperature steady when its net drift over that minute is no larger than the ordinary short-term noise (the usual up-and-down wiggles) in the trace. If a clear upward or downward drift remains, wait longer. Every recorded temperature in this part is a **steady-state** temperature.

Use a table like this:

Direction	PWM	Start Temperature ($^{\circ}\text{C}$)	Steady Temperature ($^{\circ}\text{C}$)	Time Waited (s)	Notes
Heat	0				
Heat					
Cool	0				
Cool					

Also save at least one temperature-versus-time trace for heating and one for cooling.

At this point, modify the fixed temperature y-axis used in Module 3 so small steady-state drifts are visible. Give GitHub Copilot the following prompt:

► **Suggested AI prompt: automatically scale the temperature axis**

Part 4: Plot Temperature Versus PWM

Make a graph of steady-state temperature T versus signed PWM, in which negative PWM is for cooling and positive PWM is for heating. Plot the heating and cooling measurements as separate data sets: use red for heating and blue for cooling, matching the color convention in the strip chart.

You may use Python, a spreadsheet, or another tool. The graph should show:

- red heating data,
- blue cooling data,
- labeled axes,
- units for temperature,
- a caption or short note explaining how steady state was chosen.

For each direction, estimate the **temperature susceptibility**, e.g. the slope of the curves,

$$\chi_T = \frac{dT}{du}, \quad u = \text{signed PWM}.$$

It tells you how much the steady-state temperature changes for one PWM count while the heat/cool direction is held fixed. Its units are **°C per PWM count**. Here u is positive for heating and negative for cooling. Both slopes are positive: increasing u raises the temperature on either branch. Moving farther into negative PWM lowers the temperature. The heating branch is typically steeper; a heating slope about twice the cooling slope is an experimental observation to investigate, not a required result. A simple finite-difference estimate is

$$\chi_T \approx \frac{\Delta T}{\Delta u}.$$

If the graph is not very linear, say so. The slope is still useful as a local or approximate measure of open-loop response.

Part 5: Guided Heating/Cooling Energy-Balance Analysis

Complete the measurements and safety tests in class. At home, use the graph from Part 4 to complete the following guided analysis. No additional physical measurements are required.

1. Measure The Two Slopes

From the approximately linear region of your graph, determine

$$m_h = \frac{dT_h}{du}, \quad m_c = \frac{dT_c}{du}, \quad r = \frac{m_h}{m_c}.$$

Report both slopes with units of °C per PWM count. State the PWM range used for each fit and note any visible curvature.

2. Use Steady-State Energy Balance

Recall the full object-face TEC heat-flow equation from the [Hardware discussion of the thermoelectric cooler](#):

$$\langle \dot{Q}_o \rangle = S_M T_o \langle I \rangle + \frac{1}{2} R_M \langle I^2 \rangle + K_M (T_r - T_o).$$

The subscript *o* denotes the **object face** of the TEC, which is attached to the controlled metal block; the subscript *r* denotes the **reservoir face**, which is coupled to the heat exchanger. Thus T_o and T_r are the absolute temperatures of those two faces.

The first term is Peltier transport, the second is the share of Joule heating delivered to the object face, and the third is passive conduction through the TEC. In the simplified model below, $\dot{Q}_{\text{TEC}}$ represents the two current-dependent terms. The TEC conduction term and the apparatus's other passive heat leaks are combined into the effective conductance term $-G(T - T_0)$. Do not count TEC conduction a second time inside $\dot{Q}_{\text{TEC}}$.

Let C be the thermal capacitance of the controlled object, T_0 its zero-PWM temperature, and G the effective passive thermal conductance from the object to its surroundings. Write the simplified energy balance as

$$C \frac{dT}{dt} = \dot{Q}_{\text{TEC}} - G(T - T_0).$$

At steady state, $dT/dt = 0$. The individual heat flows are generally not zero; their **sum is zero**. This is a general, model independent result. It is always true from the conservation of energy. Therefore,

$$G(T - T_0) = \dot{Q}_{\text{TEC}}.$$

Look at the above equation carefully. Enumerate all the terms and think about whether the heat is flowing into the object or out of the object for heating and cooling. Before writing the TEC heat rate, derive how PWM averages current. During one PWM period τ , let the current be I for a time $D\tau$ and zero for the remaining $(1 - D)\tau$, where $D = |u|/255$ is the nonnegative duty cycle and I is the signed on-state current (positive for heating, negative for cooling). Starting from

$$\langle I \rangle = \frac{1}{\tau} \int_0^\tau I(t) dt, \quad \langle I^2 \rangle = \frac{1}{\tau} \int_0^\tau I^2(t) dt,$$

show that

$$\boxed{\langle I \rangle = DI}, \quad \boxed{\langle I^2 \rangle = DI^2}.$$

Explain why $\langle I^2 \rangle$ is not generally equal to $\langle I \rangle^2$, and connect this distinction to your measured susceptibility. The Peltier term is proportional to $\langle I \rangle = DI$, and Joule heating is proportional to $\langle I^2 \rangle = DI^2$. For fixed on-state current, both ideal contributions are therefore linear in D , giving an approximately constant susceptibility. If one incorrectly used $\langle I^2 \rangle = \langle I \rangle^2 = D^2 I^2$, the predicted Joule term would be quadratic in duty cycle and the susceptibility would vary with D . Note that if we were not using PWM, but used a digital to analog converter that outputted a constant current whose amplitude could be linearly varied, then $\langle I^2 \rangle = \langle I \rangle^2$.

Compare the prediction for PWM control with the approximate linearity or curvature of your measured temperature-versus-PWM graph.

Near room temperature, let the positive quantities $\dot{Q}_P$ and $\dot{Q}_J$ be the Peltier and object-face Joule heat rates, in watts, when the PWM is fully on. With D the dimensionless duty cycle, the cycle-averaged heat rates are $D\dot{Q}_P$ and $D\dot{Q}_J$. The Peltier term changes sign when current reverses, whereas Joule heating does not. Define the signed duty cycle $d = u/255$, so $D = |d|$. The combined expression is

$$\dot{Q}_{\text{TEC}} = d\dot{Q}_P + |d|\dot{Q}_J.$$

On the heating branch $d > 0$; on the cooling branch $d < 0$. Thus

$$\dot{Q}_{\text{TEC},h} = d(\dot{Q}_P + \dot{Q}_J), \quad \dot{Q}_{\text{TEC},c} = d(\dot{Q}_P - \dot{Q}_J).$$

Substitute each expression into the steady-state balance and solve for $T_h(d) - T_0$ and $T_c(d) - T_0$. Differentiate with respect to the signed duty cycle d to show that

$$\frac{dT_h}{dd} = \frac{\dot{Q}_P + \dot{Q}_J}{G}, \quad \frac{dT_c}{dd} = \frac{\dot{Q}_P - \dot{Q}_J}{G}.$$

Both derivatives are positive in the cooling regime $\dot{Q}_P > \dot{Q}_J$, and the heating derivative is larger. Since $d = u/255$, each measured slope with respect to signed PWM includes a factor of $1/255$, which cancels from their ratio $r = m_h/m_c$. Show that

$$\boxed{\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{r - 1}{r + 1}}.$$

Evaluate $\dot{Q}_J/\dot{Q}_P$ using your measured value of r . As an algebra check, if $r = 2$, the result should be $\dot{Q}_J/\dot{Q}_P = 1/3$.

3. Find And Use The Laird Maximum-Current Data

Open the [Laird CP14-127-045 data sheet](#). Manufacturer data sheets contain the information needed to design with a component, but they are written for many users and can be difficult to read. Part of this exercise is deciding which entries and operating conditions apply to the class TEC.

Find the table or column for the class model at a hot-side temperature of **27 °C**. Locate and record all of the following, with units:

- module resistance, R_M ;
- maximum current, I_{max} ;
- maximum cold-side heat pumping at $\Delta T = 0$, $Q_{c,\text{max}}$; and
- maximum temperature difference, ΔT_{max} .

In one sentence each, explain what the quantity means and state the operating condition attached to it. Do not ask an AI system for the numbers before you have found them yourself. Afterward, you may give

the data sheet and your interpretation to an AI system and ask it to check whether you selected the correct values and conditions.

These are data-sheet maximum-current conditions. They are not necessarily the conditions in your apparatus when $D = 1$: full duty means that the H-bridge is continuously on, while the actual current depends on the power-supply voltage and current limit, H-bridge voltage drop, wiring, and TEC resistance.

At $\Delta T = 0$ the passive conduction term is zero. The simple symmetric TEC model assigns half of the total Joule heat to each face, so calculate the object-face Joule heat rate at the data-sheet maximum current:

$$\dot{Q}_{J,\max} = \frac{1}{2} I_{\max}^2 R_M.$$

Cooling at the object face is the Peltier heat pumping minus this Joule heat:

$$\dot{Q}_{c,\max} = \dot{Q}_{P,\max} - \dot{Q}_{J,\max}.$$

All three quantities in this equation are heat-transfer rates in watts. Use this relation to find $\dot{Q}_{P,\max}$, and then calculate the data-sheet maximum-current prediction

$$r_{\text{Laird},\max} = \frac{\dot{Q}_{P,\max} + \dot{Q}_{J,\max}}{\dot{Q}_{P,\max} - \dot{Q}_{J,\max}}.$$

4. Interpret The Comparison

Compare $r_{\text{Laird},\max}$ with your measured r . Do not assume that they should agree. State why $D = 1$ does not necessarily imply $I = I_{\max}$. Also consider PWM rather than steady DC, finite temperature differences, passive heat paths, changing material properties, and fitting a slightly curved graph with one slope.

Also answer: when the object is hotter than room temperature, which way does passive heat flow? What about when the object is colder? Explain why approximately symmetric passive conduction opposes both heating and cooling but does not by itself explain unequal slope magnitudes.

Part 6: Submit A2

A2: TEC Heating And Cooling Analysis

- **Type:** team, 10 points
- **Due:** Monday, October 5, at **6:00 PM**
- **Moodle file:** `A2_Lastname_Lastname.pdf`
- **Moodle submission:** Each student uploads the team PDF separately; teammates may upload the same PDF
- **Repository file:** none required for this short analysis

Submit a **one-to-two-page PDF** containing:

1. your Part 4 graph, showing the heating and cooling measurements as separate data sets with fitted lines over the ranges used;
2. your measured heating and cooling slopes, their units, and their ratio r ;
3. the PWM average-current proof, the steady-state slope derivation, and its numerical result;
4. the cited Laird data-sheet values, your explanation of their meanings and conditions, and the resulting calculation and predicted ratio;
5. a comparison of the measured and data-sheet ratios;
6. your answer about passive conduction; and
7. a concise conclusion of approximately 100-150 words explaining what the measurements imply about Peltier transport, Joule heating, and conduction.

Do not repeat the C2/C3 circuit sketches, apparatus descriptions, safety demonstration, or code documentation in A2. Retain the class data and working code for later modules, but no new Git checkpoint is required for this short assignment.

A2 Rubric

Criterion	Points
Part 4 graph, measured slopes, units, fitting ranges, and ratio are clearly presented	2
PWM averaging proof, steady-state energy balance, and slope-ratio derivation are correct	3
Relevant Laird values and operating conditions are correctly located, cited, interpreted, and used in a dimensionally clear calculation	2

Criterion	Points
Comparison and passive-conduction explanation show sound physical reasoning	2
PDF is concise, legible, and complete	1

C4 Oral Questions: Open-Loop TEC Calibration

These questions carry the Module 4 learning objectives into the announced C4 question bank. Prepare to answer one primary question and, when useful, one brief follow-up using your apparatus, graph, or the Laird data sheet.

1. Explain how the software temperature limit and the hardware thermal switch protect the TEC independently. How did you test the software limit without intentionally heating the apparatus to its limit?
2. Use your temperature-versus-signed-PWM graph to explain how you identified steady state and obtained the heating and cooling susceptibilities. State the units, fitting ranges, and any visible curvature.
3. Why can the heating and cooling susceptibility magnitudes differ? Explain what happens to the Peltier, Joule-heating, and passive-conduction terms when the current direction reverses.
4. Locate the required class-TEC specifications in the Laird data

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Appendix: Operational Temperature of the Safety Edit

You may ask an AI coding assistant for help, but you must test and understand the result. After using AI, identify the lines that changed and explain how the safety limit works.

► Show a possible prompt